



# HSB

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## ***Three phase system Design using Lab VIEW***

**Presented By:**

**Md Jahangir Alam – (5300396)**

**EMDADUL HOQE – (5172827)**

**Guided By:**

**Prof. Dr. Ing. Friedrich Fleischman**

**Dipl.-Ing. René Ramson**

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Md. Jahangir Alam- (5300396)  
Emdadul Hoque. – (5172827)

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# Chapter 1

## Introduction

### 1.1 What is Three phase system.

The system "three phase system" refers to a system that consists of three phases, or three wires through which current flows, with one neutral line serving to carry fault current to earth. To put it another way, a three-phase system is one that employs three wires for generating, transmission, and distribution. If the neutral wire and one of the three phase wires are removed, the three-phase system can also be operated as a single-phase system. The 3-phase system's line currents add up to zero, and their phases are discriminated at a 120-degree angle.

The four wires of a three-phase system consist of three current-carrying conductors and one neutral. The neutral conductor's cross section area is half that of the live wire. The neutral wire's current is three times the current components in the zero-phase sequence because it is equivalent to the line currents of the three wires added together.

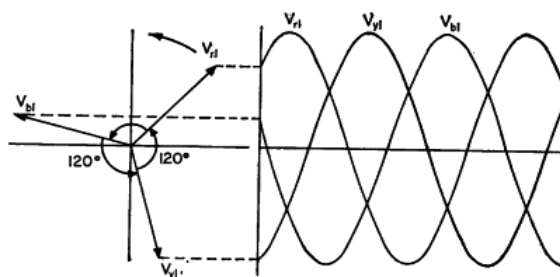


Fig 1. Three phase system

The three-phase system provides a few benefits, including the need for fewer conductors than the single-phase system. Additionally, it provides a constant supply to the load. Higher efficiency and less losses are features of the three-phase system.

The generator receives the three-phase voltage from the three-phase system, which has an identical magnitude and frequency. It offers uninterruptible power, meaning that even if one phase of the system is disrupted, the remaining two phases still supply power. The magnitude of the current in one phase of the system is equal to the total current in the other two phases. The system must operate with a **120 Degree** phase difference between its three phases. If not, the system falls

## ***1.2 key details about three-phase systems***

### ***Phase Separation:***

In terms of phase angle, the currents in the three phases are separated by 120 degrees. This phase separation enables a continuous and virtually constant flow of electricity, resulting in smoother electrical equipment performance.

### ***Voltage:***

The voltage between any two phases (line voltage) is three times (about 1.732 times) that of a single phase (phase voltage). If the phase voltage is 230 volts, the line voltage is around 400 volts.

### ***Power generating and Distribution:***

Because of its capacity to effectively transport huge amounts of power, three-phase systems are extensively utilized for power generating, transmission, and distribution. Three-phase generators produce electricity, which is subsequently converted to various voltage levels for distribution.

### ***Motor operation:***

Three-phase induction motors are extensively employed in industrial applications. They are tough, efficient, and self-starters, making them ideal for a variety of duties.

### ***Phase sequence:***

The phase sequence is the order in which the phases achieve their highest positive values. The phase sequence in a typical system is

usually A-B-C. The right phase sequence is critical for motor and equipment operation.

## ***1.3 Types of Connection***

### ***Delta ( $\Delta$ ) Connection:***

Three-phase power systems employ a particular kind of electrical connection called a delta connection, commonly referred to as a delta-wye connection or delta-delta connection. Because of its effectiveness and balanced power supply, three-phase power systems are often used in industrial applications and electrical distribution networks.

A delta connection connects the three phases in a triangle-like or delta shape. To create a closed loop, each phase is linked to the beginning of the following phase. The Greek letter delta ( $\Delta$ ), which resembles a triangle, is frequently used to symbolize this relationship. The load or electrical equipment in a delta connection is typically connected between each phase's terminals, creating the triangle's vertices.

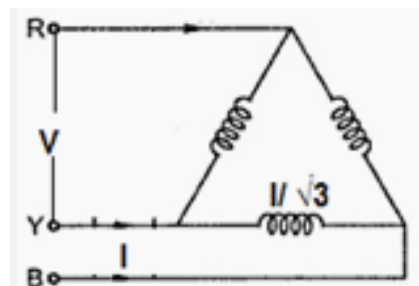


Fig 2. Delta connection

### ***The Star (Y) Connection:***

Another kind of electrical connection used in three-phase power systems is the Y connection, sometimes referred to as a wye connection or star connection. It connects three-phase loads and sources similarly to the delta connection, but it has a unique configuration and characteristics.

The three stages of a Y connection are joined at a same location, sometimes referred to as the "neutral" or "star point." The load or electrical devices are linked to the opposite ends of the phases. The name "Y connection" refers to how this arrangement resembles the letter "Y." The Greek letter "Y" or the letter "W" (for wye) can be used to denote the Y connection.

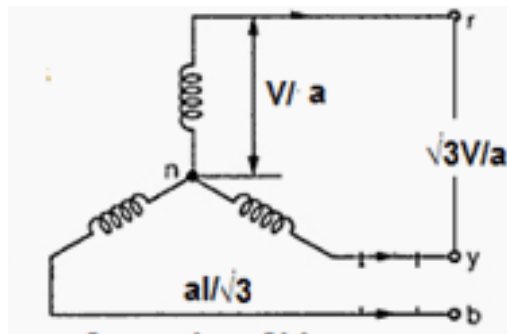


Fig 3. Y connection

## Chapter 2

### Lab VIEW

#### 2.1 Introduction of LabVIEW

LabVIEW (Laboratory Virtual Instrument Engineering Workbench) is a robust software platform created by National Instruments for a range of engineering, scientific, and industrial applications. It offers a graphical interface for programming that enables users to build virtual instruments and carry out data gathering, analysis, and control operations. The versatility and usability of LabVIEW make it popular in academia, research, and industry.

#### 2.2 Key Features of LabVIEW:

##### Graphical Programming:

Graphical Programming is one of LabVIEW's distinguishing characteristics. Instead of traditional text-based coding, LabVIEW



employs a visual technique in which you link functional blocks (VIs, or Virtual Instruments) to form your program's logic.

### **Virtual Instrumentation:**

With LabVIEW, users may create virtual instruments that imitate real-world measurement and control systems. These virtual instruments may communicate with physical devices and analyze data from a variety of sources.

### **Data Acquisition:**

LabVIEW can communicate with a broad range of hardware devices, including sensors, data acquisition cards, and instruments. This enables you to gather and analyse data from a variety of sources, making it useful for jobs such as experimental data collecting.

### **Signal Processing:**

LabVIEW has capabilities for signal processing, analysis, and visualization. Using built-in functions and modules, you may do operations like filtering, Fourier analysis, and waveform presentation.

### **User Interface Design:**

You may utilize LabVIEW to develop user-friendly graphical user interfaces (GUIs) for your applications. The drag-and-drop interface design tools allow you to alter the appearance and functionality of your program's front panel.

### **Modularity and reusability:**

LabVIEW encourages modular programming, making it simple to reuse code components across several projects. This can save time and effort when designing comparable applications.

### **Parallel Processing:**

LabVIEW enables parallel programming, enabling you to run many processes at the same time. This is especially beneficial for jobs requiring real-time control or high-performance processing.

**Integration:**

LabVIEW offers integration with various programming languages like as C/C++, .NET, and Java.

**2.3 Arduino:**

Arduino is an open-source electronics platform with simple hardware and software. Arduino boards can read inputs like a light on a sensor, a finger on a button, or a Twitter tweet and transform them into outputs like starting a motor, turning on an LED, or posting anything online. You may command your board to perform anything by passing a series of instructions to the board's microprocessor. To accomplish so, you utilize the Arduino programming language (based on Wiring) and the Arduino Software (IDE) (based on Processing).

Over the years, Arduino has served as the brain of hundreds of projects, ranging from simple household items to complicated scientific instruments. A global community of makers - students, hobbyists, artists, programmers, and professionals - has rallied around this open-source platform.



Fig 4. Arduino Mega

## ***2.4 Functionality of Arduino***

**Microprocessor:** The Arduino Mega is generally based on the ATmega2560 microprocessor. This microcontroller has 256KB of flash memory for program storage, 8KB of RAM, and 4KB of EEPROM for data storage.

**Digital and Analog I/O Pins:** The Arduino Mega has a large number of digital and analog input/output pins. It has 54 digital I/O pins, 15 of which may be utilized as PWM (Pulse Width Modulation) outputs, and 16 analog input pins.

**Power Supply:** The Arduino Mega may be supplied through USB, an external DC power supply, or an optional power-over-Ethernet (PoE) module.

**Voltage Regulation:** The board features voltage regulation to give a consistent supply voltage to the microcontroller and other components.

**Operating Voltage:** The Arduino Mega's normal operating voltage is 5V. Some variants of the PCB may also support 3.3V operation.

**Communication Interfaces:** The board has many communication interfaces, including a serial UART, a serial peripheral interface (SPI), and an inter-integrated circuit (I2C).

**Clock Speed:** The Arduino Mega's ATmega2560 microprocessor has a clock speed of 16 MHz.

**Compatibility:** The Arduino Mega works with the Arduino IDE and employs the same programming language and development environment as previous Arduino boards.

**Shields:** The Mega, like other Arduino boards, may employ shields to enhance its capabilities. Due to its bigger size, it may be incompatible with a different set of shields than smaller Arduino variants.

# Chapter 3

## Three phase system Design using Lab VIEW

### 3.1 Block Diagram Input part:

First, the input section's input voltage is set at 5 volts. In order to generate 16A ( $5 \times 35 = 16A$ ), we multiply by 3.5. Next, we use a multiplexer to turn each of the three variables—i1, i2, and i3—into a constant. To create a second phase variable constant, we use a second multiplexer and Structure Loop.

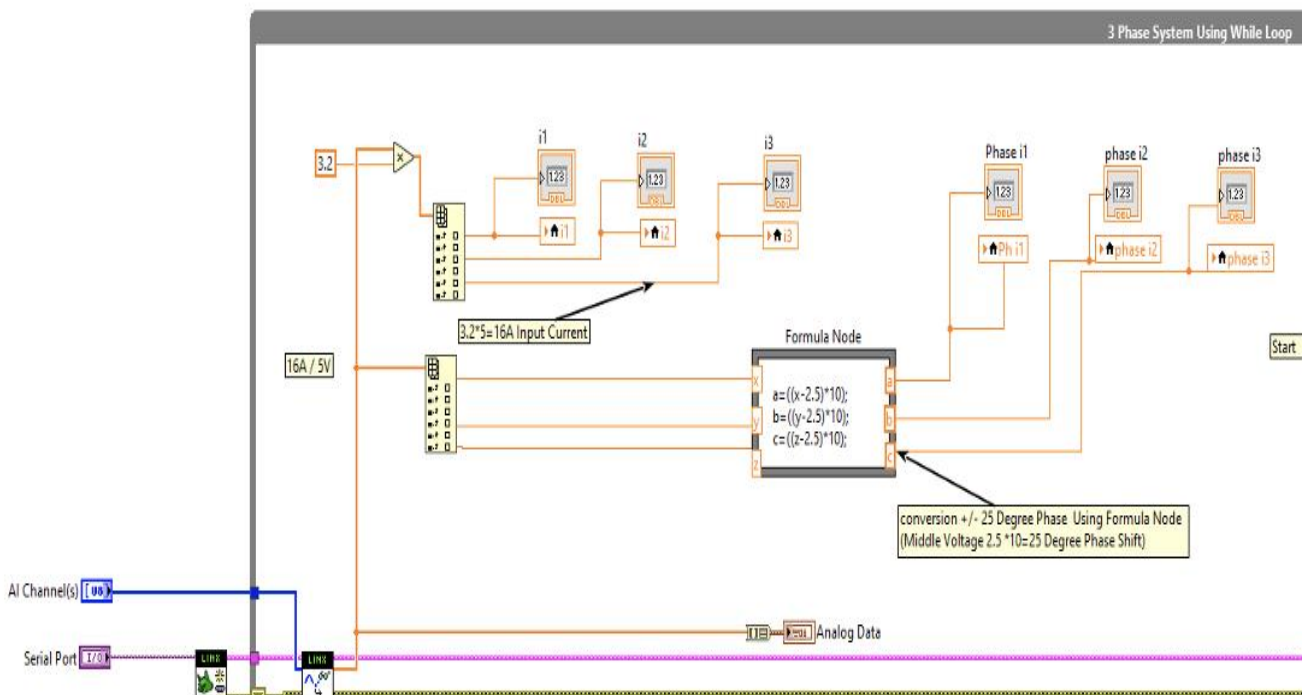


Fig 5. Block Diagram Input part

Here is the equation we use.

$$\text{Output value} = (\text{input value} - 2.50) * 10$$

Here,

x,y and z are Input value and a,b,c are output value.

Equation,

$$\begin{aligned} a &= ((x - 2.5) * 10); \\ &= (5 - 2.5) * 10 \\ &= 25 \text{ Degree Phase shift} \end{aligned}$$

so, we get output value sequentially  $b=25$  degree and  $C=25$  degree.

Three variable currents with a combined value of 16A were found on the input side( $i_1, i_2, i_3$ ). Additionally, get phase constants  $\text{phase}_1$ ,  $\text{phase}_2$ , and  $\text{phase}_3$ , and ensure that their values correspond to a 25-degree phase shift.

### 3.2 Oscilloscope:

This section applies a three-input sinusoidal signal arrangement. In this case, we apply a 50HZ frequency and a 120-degree phase shift from one signal to another. The signal phase degrees are 0, 120 and 240. Then we employ three sinusoidal signals with phase and current variable constants as input variables. This signal also employs phase shift. After we multiply these signals by 12, we obtain 192V, which is close to 220. We divide the signal and display it on the oscilloscope.

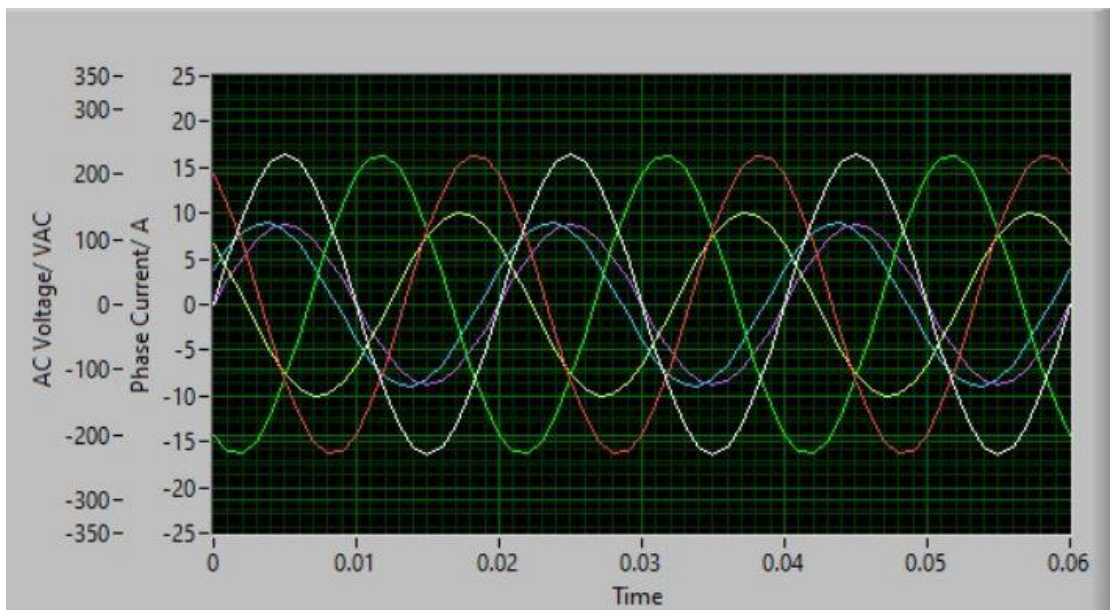


Fig 6. Oscilloscope

### 3.3 Three Phase Meter Display:

Display the signal as a current and voltage scale on the oscilloscope. This section displays six signals: current  $i_1$ ,  $i_2$ ,  $i_3$  and voltage  $u_1, u_2, u_3$ . When we increase the current using the Arduino, the graph shows a different value in the 120-degree phase shift.

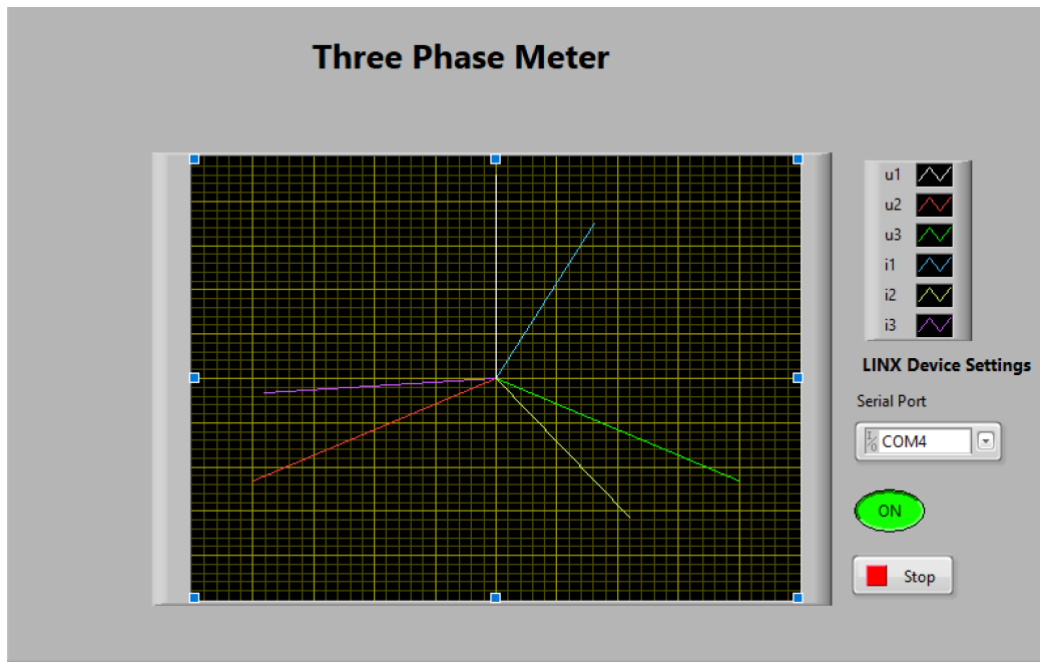


Fig 7. Front panel Three phase meter

Then we separated the signal and used measurement instruments to measure u1, u2, u3, and i1, i2, i3. Converts dynamic data from measurement instruments and displays it in a variable constant function. Subtract the signal, then show in constant (Ph i1 m, Ph i2 m, Ph i3 m) after that we convert the signal from degree to radiant conversation.

### 3.4 Conversation of signal using case structure and Loop Formula:

The signal was then separated and measured using measurement tools measurement u1, measurement u2, measurement u3, and measurement i1, measurement i2, measurement i3.



Fig 8. Conversation of signal using case structure and Loop Formula

Dynamic data from measuring instruments is converted and displayed in a variable constant function. Subtract the signal from the constant (Ph i1 m, Ph i2 m, Ph i3 m), and then convert the signal from degree to radiant conversation.

Formula loop  $a = (((w-180)/360) * (2 * \pi));$   
 $b = (((x-180)/360) * (2 * \pi));$   
 $c = (((y+120)-180)/360) * (2 * \pi);$   
 $d = (((z+240)-180)/360) * (2 * \pi);$

In this equation w,x,y z are input and a,b,c,d are output signal.  
 After radiant conversation, we use the negative sine and cosine signals to multiplex the signal. We transform the signal vector (v1,v2,v3, and i1,i2,i3) using a binary array and bundle. Using a bundle Array, the entire signal is shown in a three-phase meter.

### 3.5 Stop Button and Start Button:

In this section, we utilize a Boolean function to construct a stop button so that we may manage the program without using the run button. Here are three stop buttons that were made with the use of component arithmetic and/or a gate (the inputs were data type of weir, which we used in a for loop, and another was compound math).

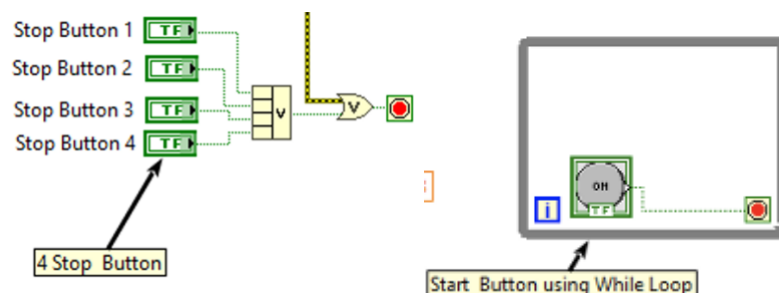


Fig 9. Start and Stop Button

While loop is used to build a start button. To control the entire function without a run button, we utilize boolean as a button in this start button.

### 3.6 Readings:

The Arduino pin diagram, which is 0,1,2,3,4,5, was initially announced in the readings section. When using measurement instruments, the u/v value is nearly fixed, and we fix the value as input data. This section displays the entire output value, and the phase angle between the input signal's left and right sides is 90 degrees, giving a total angle of 120

degrees. The Arduino knob may be used to increase and reduce the current to a maximum of 16A.

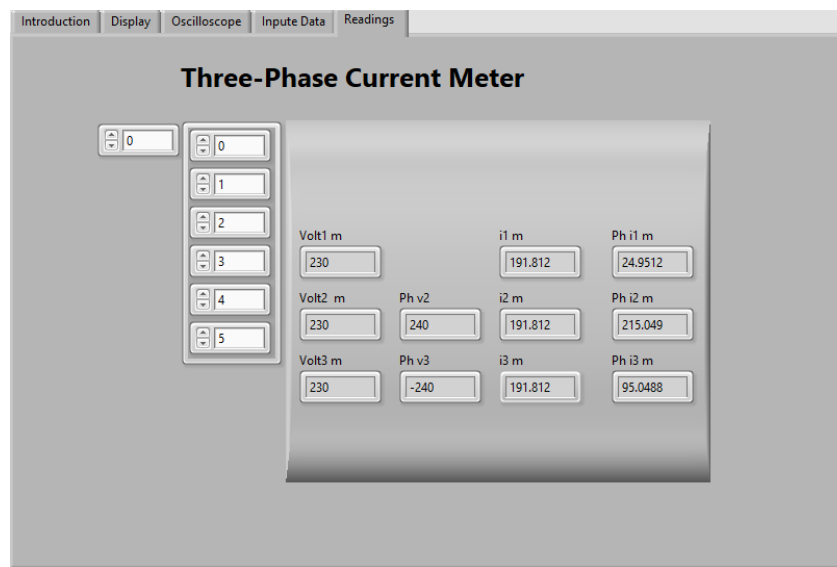


Fig 10. Three Phase current meter Readings

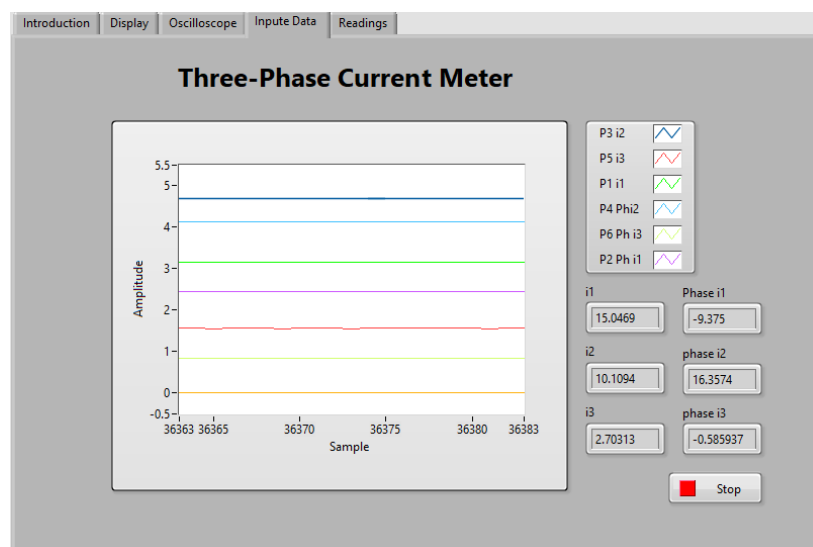


Fig 11. Three Phase current meter Input Data



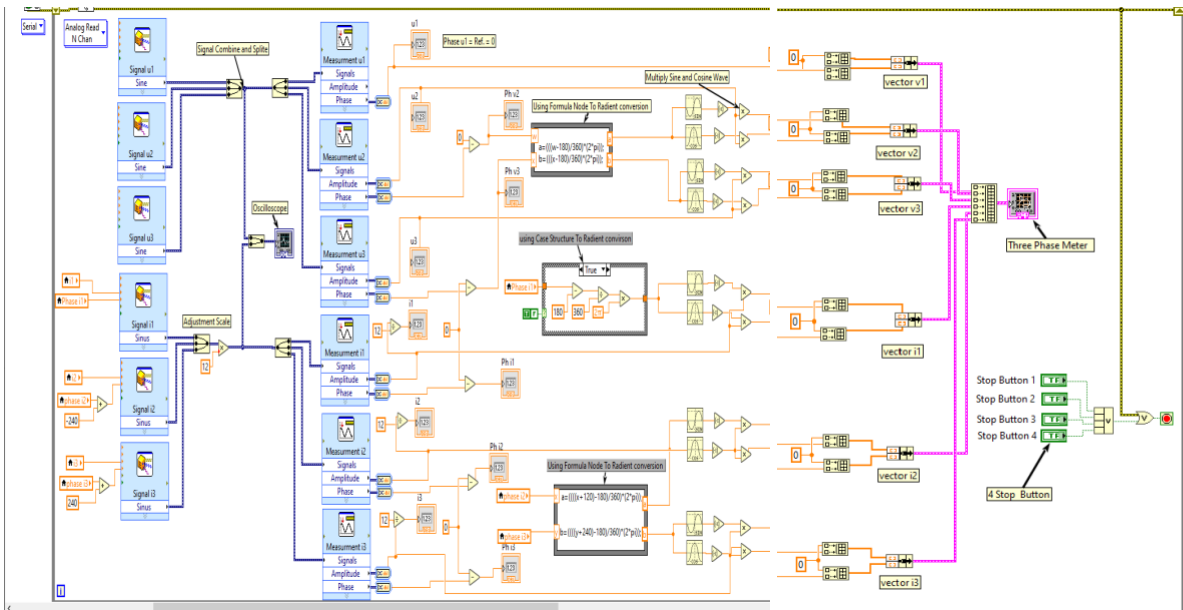


Fig 11. Block Diagram of this project

## Chapter 4

### Conclusion

In this paper, we investigated the use of LabVIEW to create a basic three-phase system simulation. We successfully simulated the essential properties of a three-phase electrical system by creating three sinusoidal impulses with a 120-degree phase difference. This simulation was used to demonstrate the basic ideas underpinning three-phase systems.

The LabVIEW platform made signal creation, manipulation, and visualization simple. We were able to dynamically alter frequency and amplitude, see the combined waveform, and even visualize the individual phases thanks to the integration of controls, signal production tools, and waveform charts.

It should be noted, however, that real-world three-phase systems can be far more complex. The behavior is heavily influenced by factors such as phase angles, voltage levels, harmonic content, and load balance.

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